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PART I

Study 2 Results

1

Overview of the registered outcome

All 1,341 registered runs completed between 18 Oct 05:07 and 19 Oct 08:28 IST. None failed, needed a retry, or diverged; every run was trained from the registration commit; every result matched its registered split hash; and the collector reported no missing, duplicated, or mismatched task. The registered analysis (`analysis/study2/run_all.sh`) ran once, on 19 October at 12:56.

Table 1.1: The five registered Study 2 tests (H5 in pp of gap per pp of recoverability).

	Question	Estimate	Adjusted p	Verdict
H1	SIPaKMeD P1 − P2 accuracy (primary, tested alone)	0.97	<0.001	**supported**
H2	CRIC P1 − P2 accuracy	12.08	<0.001	**supported**
H3	Leakage at matched diversity (2C)	1.73	0.02	**supported**
H4	Own-context route through backgrounds (2B)	0.42	0.23	**not supported**
H5	Gap versus suppressed recoverability (2A)	0.06	0.16	**not supported**

Gate A (H1 supported) means the inflation is reported as certified. Gate B was "murky": H3 was supported but H4 was not and H5 was not supported (one of two architectures failed its manipulation check). That is the result that motivated Amendment 8.

2

H1: crop-level splitting inflates SIPaKMeD accuracy

2.1 The registered test

Mean test accuracy was 97.14% under P1 and 96.18% under P2. The registered residual check flagged heavy tails (residual skewness −0.96, excess kurtosis 3.64, Shapiro–Wilk p <0.001), so, as registered, the two-way cluster bootstrap is the primary inference: 0.97 pp (95% interval 0.59–1.39, bootstrap p <0.001). Every sensitivity analysis agrees (Table 2.1): the mixed model gives the same estimate with Satterthwaite $df = 38.8$ and p <0.001 under Kenward–Roger, the seed-level sign-flip test gives p <0.001, and the binomial mixed model a log odds ratio of 0.304 (p <0.001).

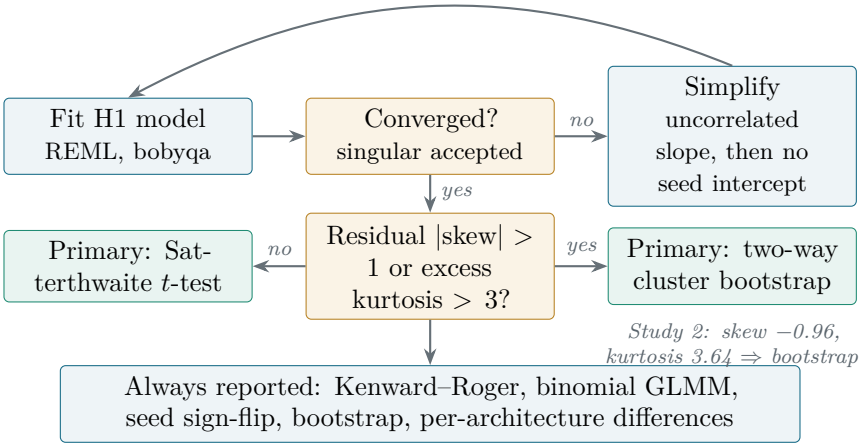


Figure 2.1: The registered H1 decision path. The residual check fired, so the cluster bootstrap became the primary inference exactly as registered; every other method gave the same answer.

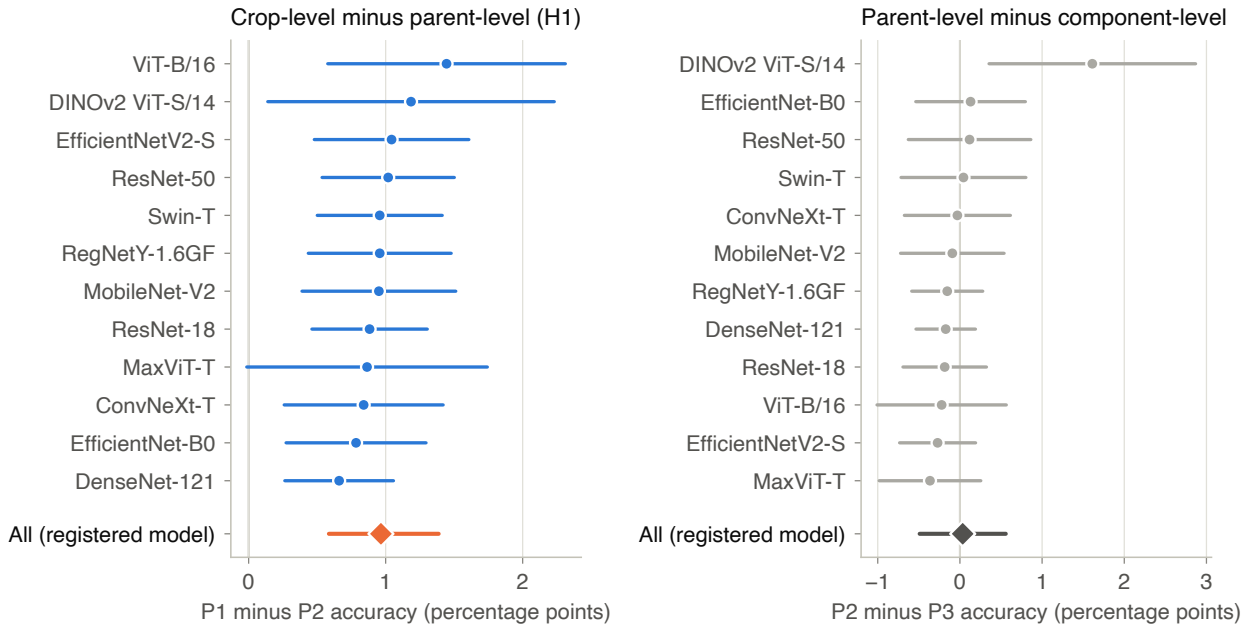


Figure 2.2: P1 − P2 (left) and P2 − P3 (right) per architecture (mean and 95% t interval over 20 seeds) with the registered pooled estimates.

Table 2.1: Every protocol model with its registered sensitivity analyses: Satterthwaite and Kenward–Roger degrees of freedom, two-way cluster bootstrap, and seed sign-flip test. The primary source is the bootstrap when the registered residual check fails.

Contrast	Model	Est.	SE	df_S	p_S	df_{KR}	p_{KR}	Bootstrap CI	Sign-flip p	Primary
H1 accuracy	full	0.97	0.17	38.8	<0.001	18.8	<0.001	0.59–1.39	<0.001	cluster_bootstrap
H1 macro-F1	full	0.96	0.17	38.8	<0.001	18.7	<0.001	0.59–1.39	<0.001	cluster_bootstrap
H1 balanced accuracy	full	0.98	0.17	38.8	<0.001	18.7	<0.001	0.60–1.40	<0.001	cluster_bootstrap
P2 − P3 accuracy	full	0.03	0.25	25.7	0.90	25.2	0.90	−0.45–0.60	0.88	lmm
H2 CRIC accuracy	full	12.08	0.40	8.8	<0.001	8.8	<0.001	11.23–12.92	0.002	lmm
CRIC P1′ − P2	full	12.11	0.43	17.5	<0.001	10.1	<0.001	11.21–13.05	0.002	lmm
CRIC balanced accuracy	full	19.64	0.71	18.0	<0.001	7.2	<0.001	18.00–21.34	0.002	lmm
CRIC macro-F1	full	20.59	0.60	6.3	<0.001	6.3	<0.001	19.19–22.02	0.002	lmm

2.2 Consistency

All 12 architectures scored higher under P1 (0.66–1.44 pp) and 11 of them individually had intervals excluding zero. At the seed level the picture is noisier but one-sided (Figure 2.4). The parent-level error rate is 1.34 times the crop-level one. Macro-F1 (0.96 pp) and balanced accuracy (0.98 pp) moved by the same amount.

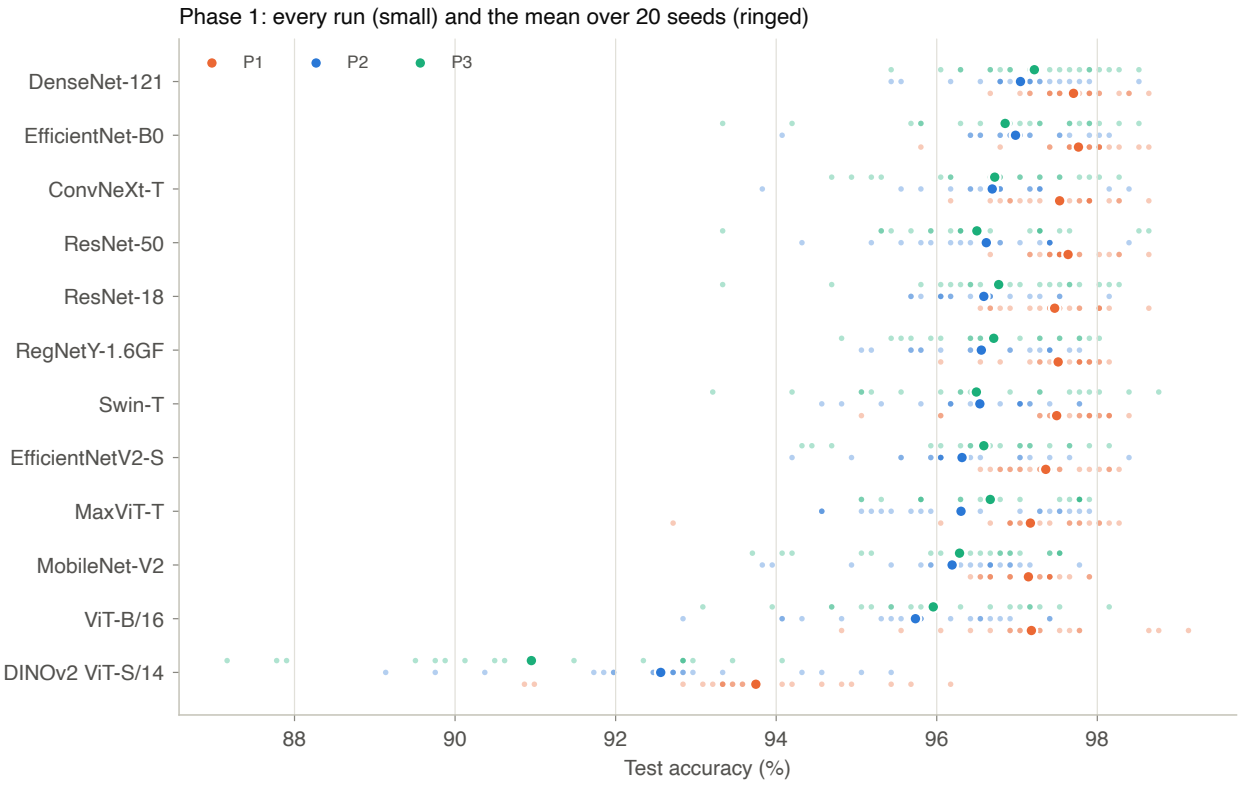


Figure 2.3: Every Phase 1 run and the per-protocol mean over 20 seeds, by architecture.

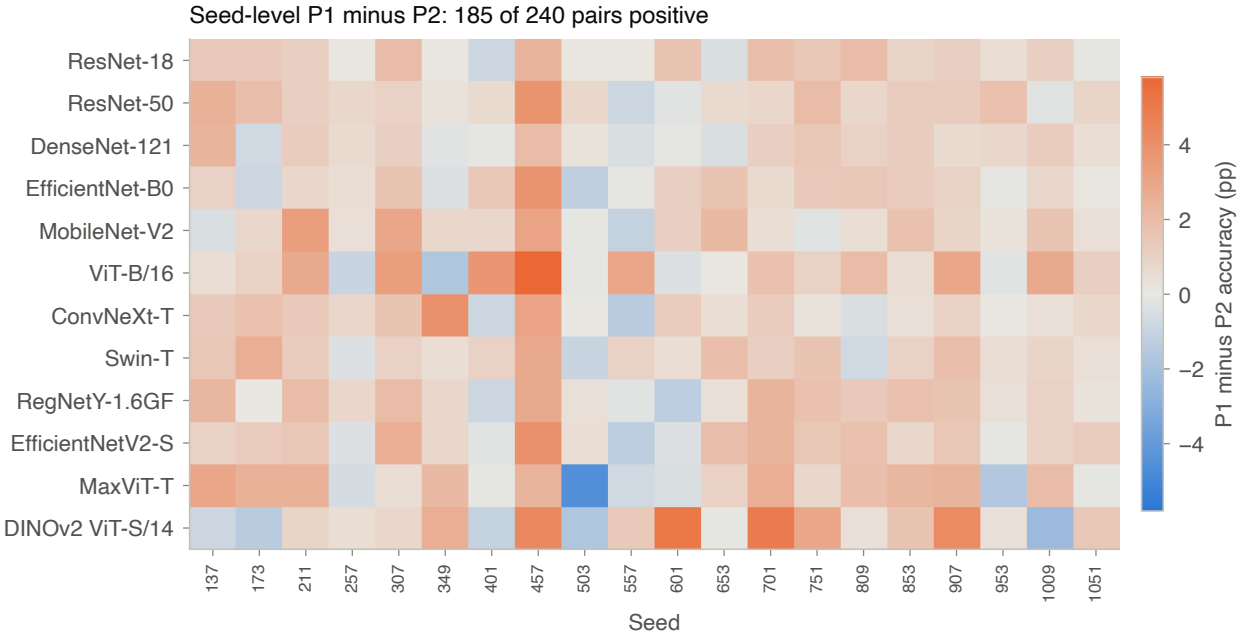


Figure 2.4: P1 − P2 accuracy for every architecture and seed (diverging scale, zero in grey).

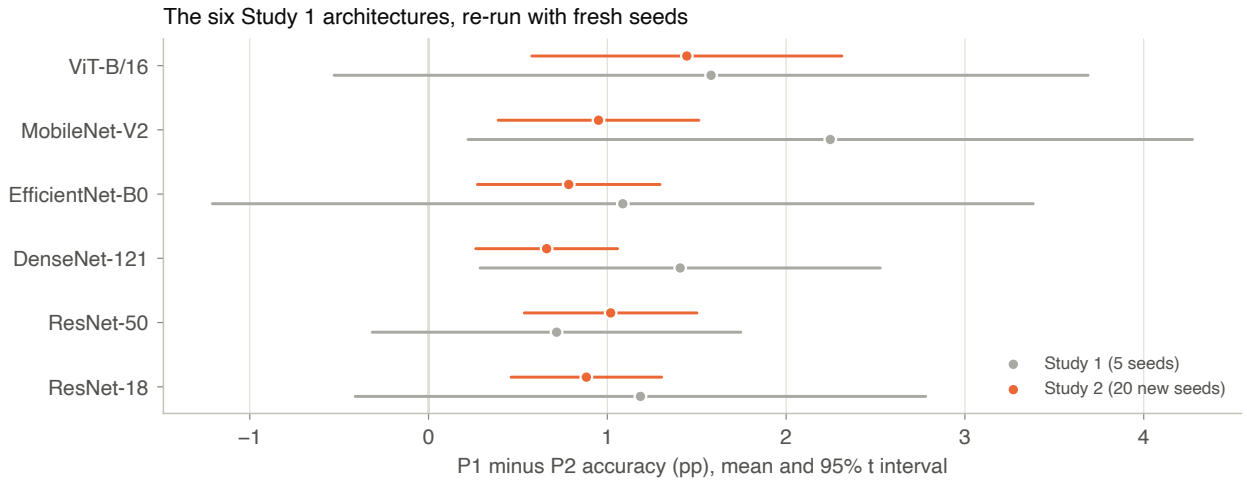


Figure 2.5: The six Study 1 architectures: Study 1's five-seed estimates and Study 2's twenty-seed estimates of P1 − P2. Study 2's intervals are narrower and all exclude zero.

2.3 P2 versus P3, rankings, and the ordering gate

Merging near-duplicate-linked parents on top of parent-level splitting changed nothing (P2 − P3 0.03 pp, −0.49 to 0.56, $p = 0.90$). The registered P3 ≤ P2 ≤ P1 ordering passed for 4 of twelve architectures, as expected when two protocols are equivalent and the order between them is decided by noise. The architecture ranking was largely preserved between P1 and P2 (Kendall $\tau = 0.82$, 0.42–0.86; 6/66 pairs reordered).

2.4 Crop-level evidence

Within crops that were test crops under both protocols in different seeds (3,664 crops), accuracy was 1.29 pp higher under P1 (1.10–1.50; M1). Within P1, a crop sharing source pixels with a training crop was more often correct (log odds 0.45, p <0.001), as was a crop with more training siblings (log odds 0.15 per log sibling, $p = 0.01$), adjusting for class and the crop's difficulty under P2 (M2).

2.5 Training behaviour

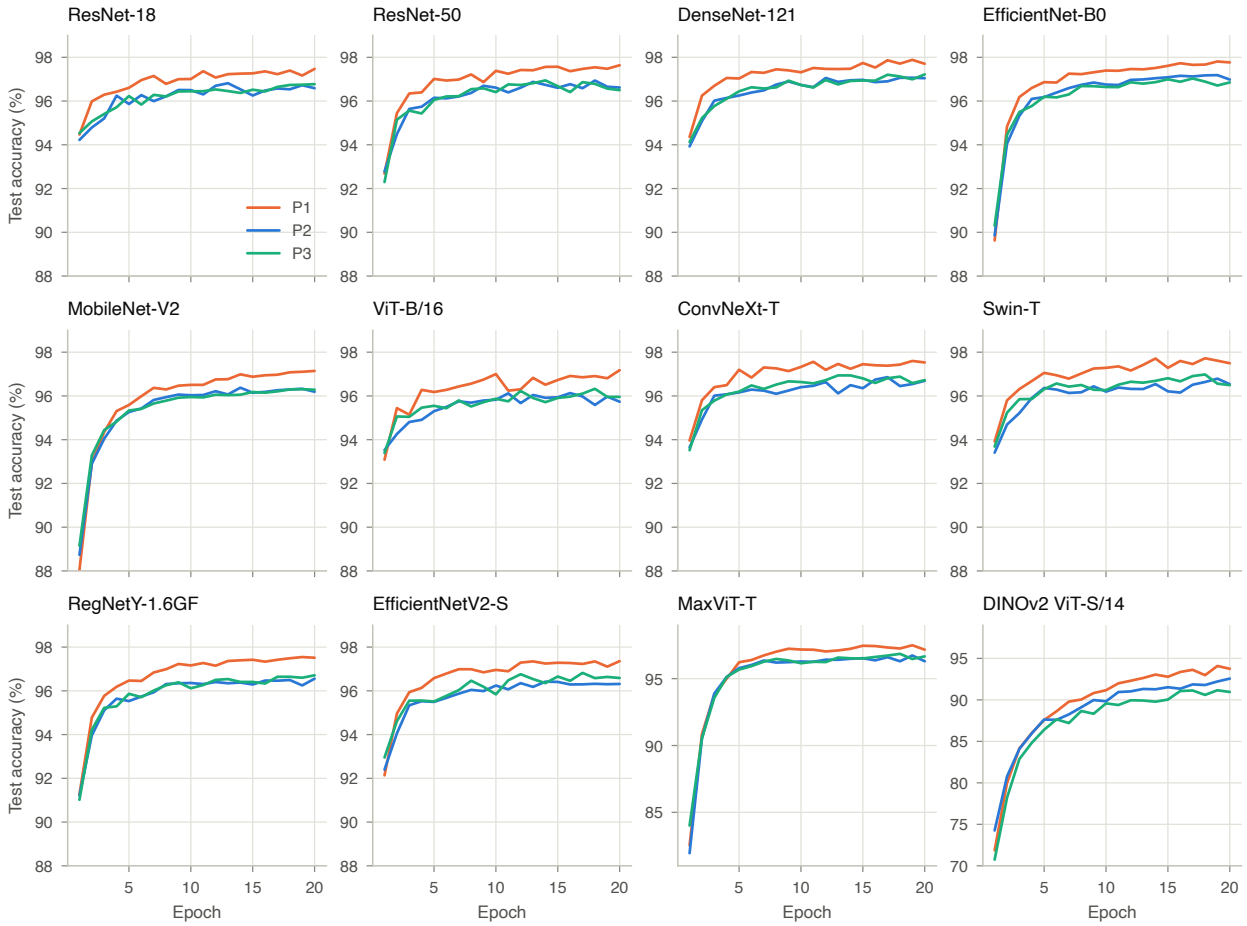


Figure 2.6: Mean test accuracy per epoch under P1, P2, and P3 (20 seeds each). The P1 advantage appears within the first few epochs and persists to the final-epoch endpoint.

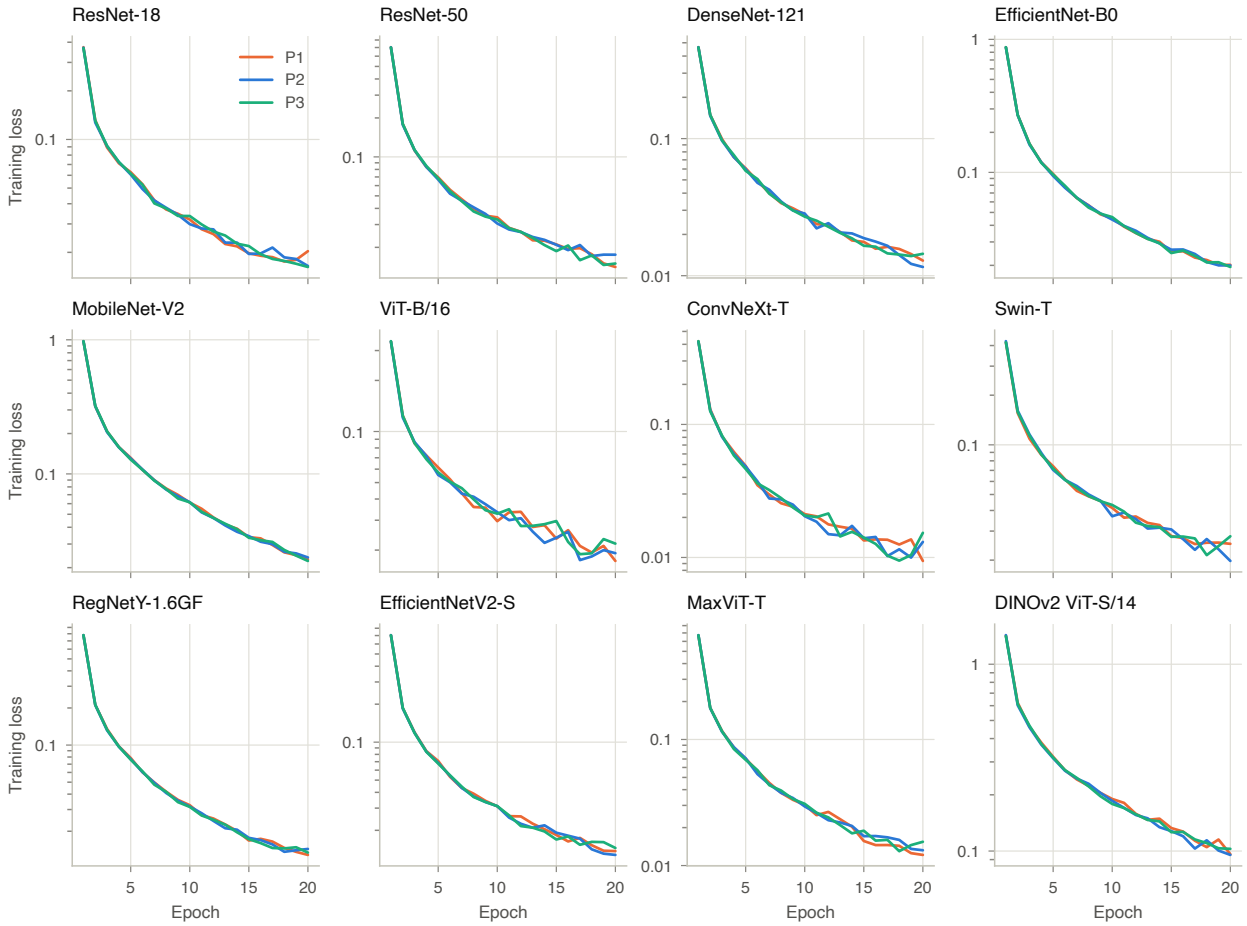


Figure 2.7: Mean training loss per epoch (log scale). Training is indistinguishable across protocols; the difference is in what the test set contains, not in how training proceeds.

2.6 Calibration and errors

Pooled over runs, the 15-bin expected calibration error was 1.78% under P1, 2.43% under P2, and 2.37% under P3, and the negative log-likelihood 0.102, 0.142, and 0.136 nats: crop-level splits flatter calibration as well as accuracy.

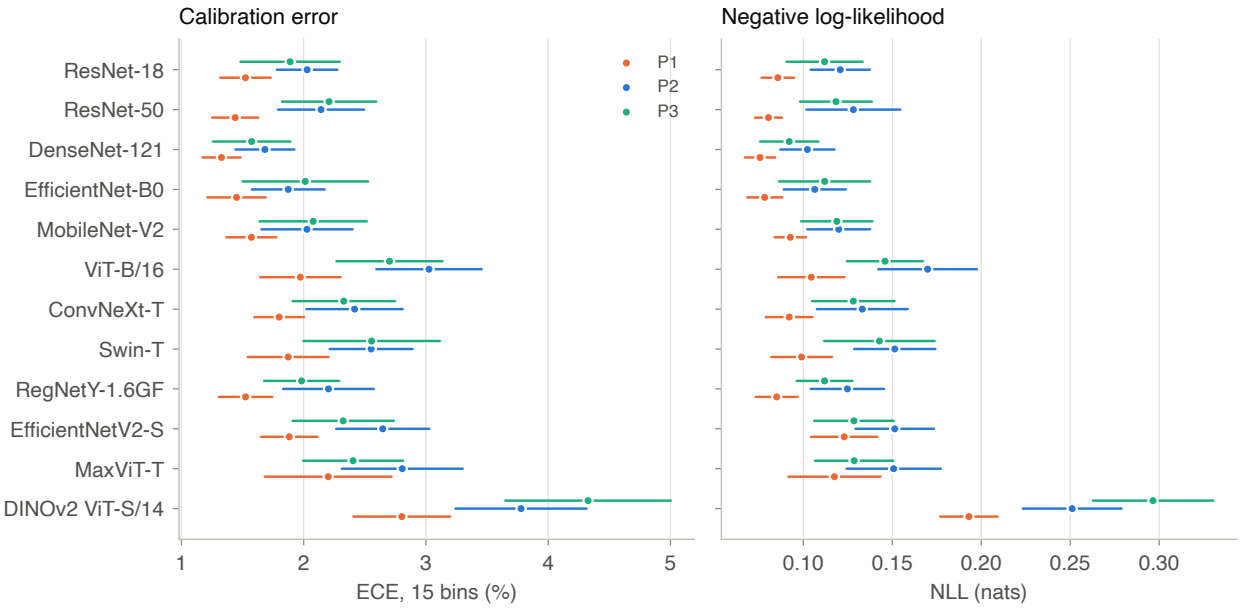


Figure 2.8: Calibration error and negative log-likelihood by architecture and protocol (mean and 95% t interval).

Table 2.2: Phase 1 (SIPaKMeD, 20 seeds per cell): test metrics in percent (mean $\pm$ SD over seeds); NLL in nats.

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
ResNet-18	P1	97.47 ± 0.56	97.48 ± 0.56	97.49 ± 0.56	1.52 ± 0.44	0.086
	P2	96.59 ± 0.68	96.60 ± 0.67	96.59 ± 0.68	2.03 ± 0.53	0.121
	P3	96.77 ± 1.22	96.78 ± 1.21	96.77 ± 1.21	1.89 ± 0.87	0.112
ResNet-50	P1	97.64 ± 0.46	97.65 ± 0.45	97.65 ± 0.46	1.44 ± 0.40	0.080
	P2	96.62 ± 0.94	96.63 ± 0.94	96.62 ± 0.94	2.14 ± 0.75	0.128
	P3	96.50 ± 1.23	96.51 ± 1.23	96.49 ± 1.26	2.21 ± 0.82	0.118
DenseNet-121	P1	97.70 ± 0.51	97.71 ± 0.51	97.72 ± 0.51	1.33 ± 0.33	0.076
	P2	97.04 ± 0.74	97.05 ± 0.74	97.05 ± 0.74	1.68 ± 0.52	0.102
	P3	97.22 ± 0.83	97.23 ± 0.83	97.21 ± 0.83	1.57 ± 0.68	0.092
EfficientNet-B0	P1	97.77 ± 0.61	97.77 ± 0.62	97.78 ± 0.61	1.45 ± 0.51	0.078
	P2	96.98 ± 0.86	96.98 ± 0.87	96.99 ± 0.85	1.87 ± 0.64	0.106
	P3	96.85 ± 1.35	96.86 ± 1.34	96.86 ± 1.34	2.01 ± 1.10	0.112
MobileNet-V2	P1	97.14 ± 0.46	97.15 ± 0.46	97.16 ± 0.46	1.57 ± 0.44	0.093
	P2	96.19 ± 1.01	96.20 ± 1.01	96.19 ± 1.02	2.03 ± 0.80	0.120
	P3	96.28 ± 1.22	96.29 ± 1.22	96.28 ± 1.22	2.08 ± 0.94	0.119
ViT-B/16	P1	97.18 ± 1.03	97.19 ± 1.02	97.19 ± 1.04	1.97 ± 0.70	0.104
	P2	95.73 ± 1.21	95.73 ± 1.23	95.74 ± 1.21	3.02 ± 0.92	0.170
	P3	95.96 ± 1.30	95.96 ± 1.31	95.97 ± 1.30	2.70 ± 0.93	0.146
ConvNeXt-T	P1	97.53 ± 0.64	97.54 ± 0.63	97.54 ± 0.64	1.80 ± 0.43	0.092
	P2	96.69 ± 0.95	96.70 ± 0.94	96.69 ± 0.95	2.42 ± 0.84	0.133
	P3	96.72 ± 1.06	96.72 ± 1.07	96.73 ± 1.05	2.33 ± 0.89	0.128
Swin-T	P1	97.49 ± 0.84	97.50 ± 0.84	97.51 ± 0.84	1.87 ± 0.71	0.099
	P2	96.54 ± 0.89	96.55 ± 0.88	96.55 ± 0.88	2.55 ± 0.72	0.151
	P3	96.49 ± 1.46	96.50 ± 1.47	96.50 ± 1.46	2.56 ± 1.19	0.143
RegNetY-1.6GF	P1	97.51 ± 0.54	97.52 ± 0.54	97.53 ± 0.54	1.52 ± 0.47	0.085
	P2	96.56 ± 0.83	96.57 ± 0.84	96.56 ± 0.83	2.20 ± 0.79	0.125
	P3	96.71 ± 0.93	96.72 ± 0.94	96.71 ± 0.93	1.98 ± 0.66	0.112
EfficientNetV2-S	P1	97.36 ± 0.54	97.37 ± 0.54	97.37 ± 0.53	1.88 ± 0.49	0.123
	P2	96.31 ± 0.95	96.32 ± 0.96	96.32 ± 0.95	2.65 ± 0.81	0.151
	P3	96.59 ± 1.09	96.59 ± 1.09	96.59 ± 1.09	2.32 ± 0.88	0.128
MaxViT-T	P1	97.17 ± 1.19	97.17 ± 1.20	97.19 ± 1.18	2.20 ± 1.11	0.117
	P2	96.30 ± 1.21	96.31 ± 1.20	96.31 ± 1.21	2.81 ± 1.06	0.151
	P3	96.67 ± 0.97	96.67 ± 0.98	96.68 ± 0.97	2.40 ± 0.87	0.129

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
DINOv2 ViT-S/14	P1	93.75 ± 1.34	93.78 ± 1.33	93.79 ± 1.33	2.80 ± 0.85	0.193
	P2	92.56 ± 1.60	92.61 ± 1.55	92.61 ± 1.60	3.78 ± 1.15	0.251
	P3	90.95 ± 1.95	90.99 ± 1.92	90.99 ± 1.95	4.33 ± 1.45	0.297

3

H2: the CRIC inflation is an order of magnitude larger

CRIC accuracy fell from 90.48% under crop-level splitting to 78.40% under slide-level splitting: H2 = 12.08 pp (11.16–12.99; Holm p <0.001). The error rate rose 2.27-fold. Because the classes are imbalanced, balanced accuracy fell further, from 78.53% to 58.89% (19.64 pp), and macro-F1 by 20.59 pp. The size-matched P1-prime gave the same inflation (12.11 pp), so it is not a training-size effect.

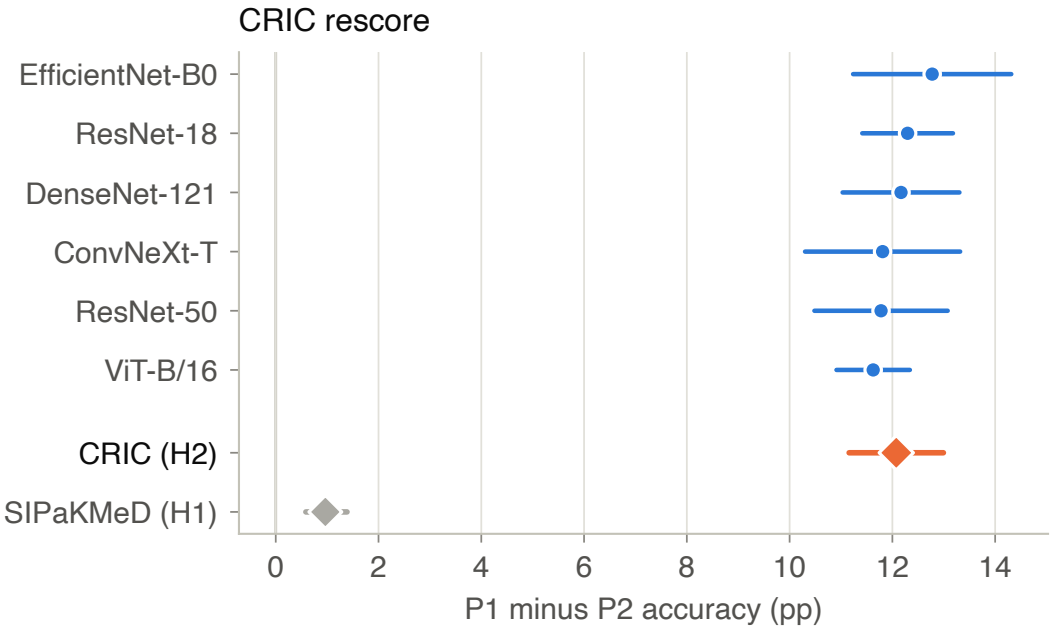


Figure 3.1: CRIC P1 − P2 per architecture, the registered estimate, and the SIPaKMeD estimate for scale.

The loss is concentrated in the abnormal classes (Figure 3.2). Recall for ASC-H, LSIL, and HSIL falls by tens of points when slides are held out, while the majority NILM class barely moves; the crop-level split had been rewarding the recognition of slides whose abnormal cells were already seen.

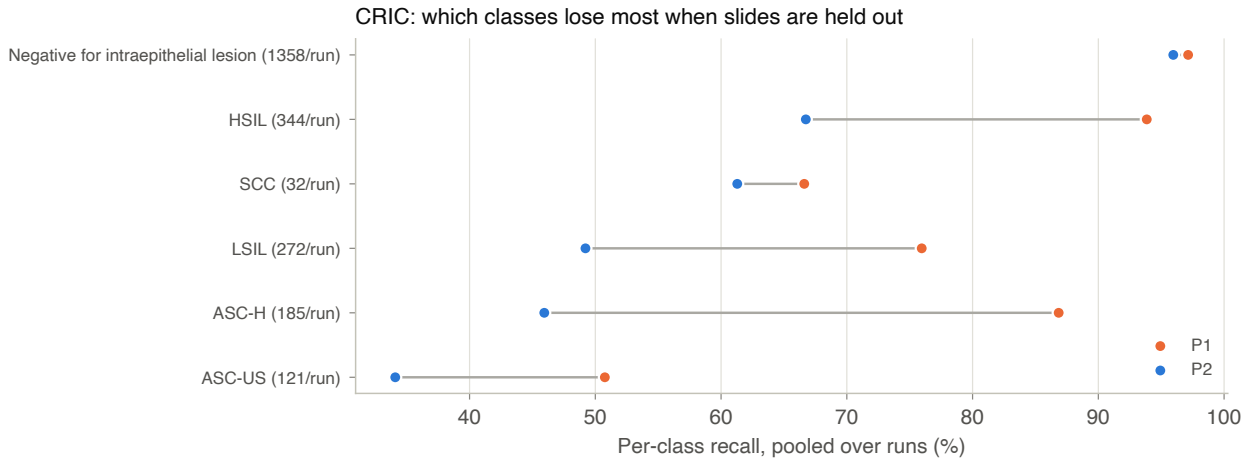


Figure 3.2: CRIC per-class recall under P1 and P2, pooled over all runs, with the mean number of test cells per class.

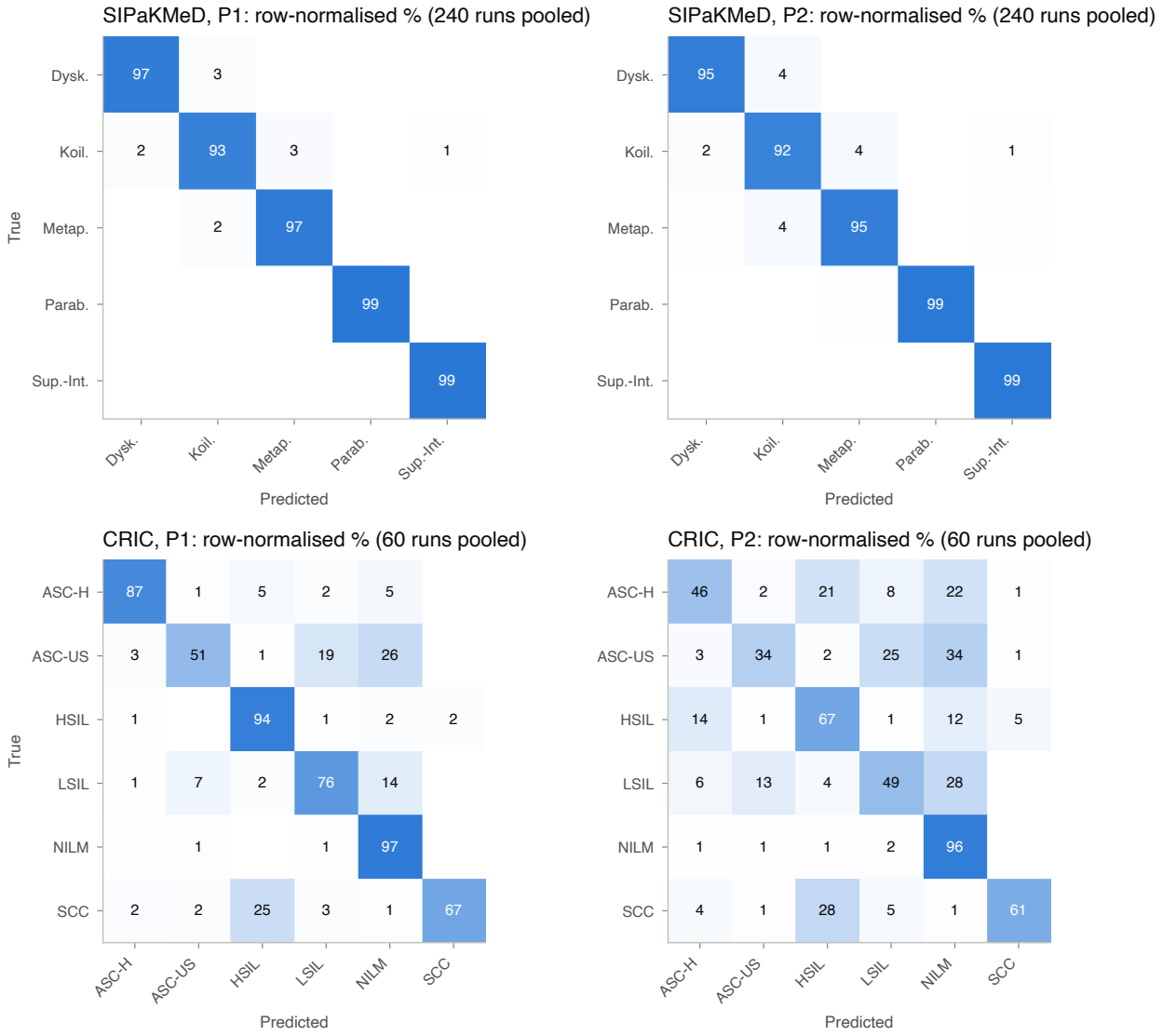


Figure 3.3: Row-normalised confusion matrices pooled over runs. SIPaKMeD errors barely change between protocols; CRIC's abnormal classes collapse toward NILM and toward each other under slide-level splitting.

Table 3.1: CRIC (10 seeds per cell): test metrics in percent (mean ± SD).

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
ResNet-18	P1	89.91 ± 0.43	79.09 ± 1.31	77.60 ± 2.07	5.50 ± 0.40	0.383
	P1-prime	89.92 ± 0.68	78.94 ± 1.43	77.42 ± 2.25	5.48 ± 0.67	0.387
	P2	77.62 ± 1.14	57.25 ± 1.45	56.79 ± 2.63	14.98 ± 1.11	1.014
ResNet-50	P1	90.54 ± 0.48	80.48 ± 1.63	79.42 ± 2.41	5.85 ± 0.33	0.388
	P1-prime	90.85 ± 0.62	81.01 ± 1.66	79.85 ± 2.02	5.41 ± 0.45	0.371
	P2	78.76 ± 1.80	60.48 ± 3.30	60.34 ± 2.84	14.97 ± 1.82	0.995
DenseNet-121	P1	90.40 ± 0.74	79.66 ± 1.93	77.97 ± 2.71	5.42 ± 0.68	0.374
	P1-prime	90.71 ± 0.48	80.22 ± 1.41	78.32 ± 1.76	5.36 ± 0.46	0.365
	P2	78.23 ± 1.50	59.08 ± 2.36	58.82 ± 2.77	14.95 ± 1.28	1.017
EfficientNet-B0	P1	90.88 ± 0.53	80.74 ± 0.99	79.78 ± 1.46	5.33 ± 0.43	0.370
	P1-prime	90.95 ± 0.60	80.95 ± 1.05	80.83 ± 1.64	5.21 ± 0.54	0.363
	P2	78.10 ± 2.15	58.73 ± 2.78	59.86 ± 2.19	14.90 ± 1.74	1.021
ViT-B/16	P1	89.83 ± 0.63	78.01 ± 2.30	76.37 ± 3.22	6.54 ± 0.95	0.436
	P1-prime	89.54 ± 0.70	78.28 ± 1.81	76.45 ± 2.95	6.55 ± 0.68	0.429
	P2	78.20 ± 1.13	59.54 ± 1.47	57.62 ± 2.28	16.14 ± 1.29	1.054
ConvNeXt-T	P1	91.29 ± 0.63	81.80 ± 1.57	80.03 ± 2.69	5.91 ± 0.58	0.392
	P1-prime	91.10 ± 0.40	80.73 ± 1.97	78.85 ± 2.17	6.02 ± 0.53	0.392
	P2	79.48 ± 1.78	61.15 ± 3.27	59.90 ± 3.99	15.82 ± 1.76	1.082

4

H3: the inflation is leakage, not training diversity

With crop count, field count, and allocation rule held equal, adding the test fields' sibling crops to training raised accuracy on the common test set by 1.73 pp (0.95–2.51; Holm $p = 0.02$). Every architecture showed it at both diversity levels (Figure 4.2). Increasing the number of distinct training fields from 504 to 720 without leakage did nothing (0.00 pp, −0.65 to 0.64), and the leak effect did not depend on diversity (interaction −0.20 pp, $p = 0.60$). The registered approximate decomposition assigns the Phase 1 gap entirely to leakage (diversity component 0.00 pp).

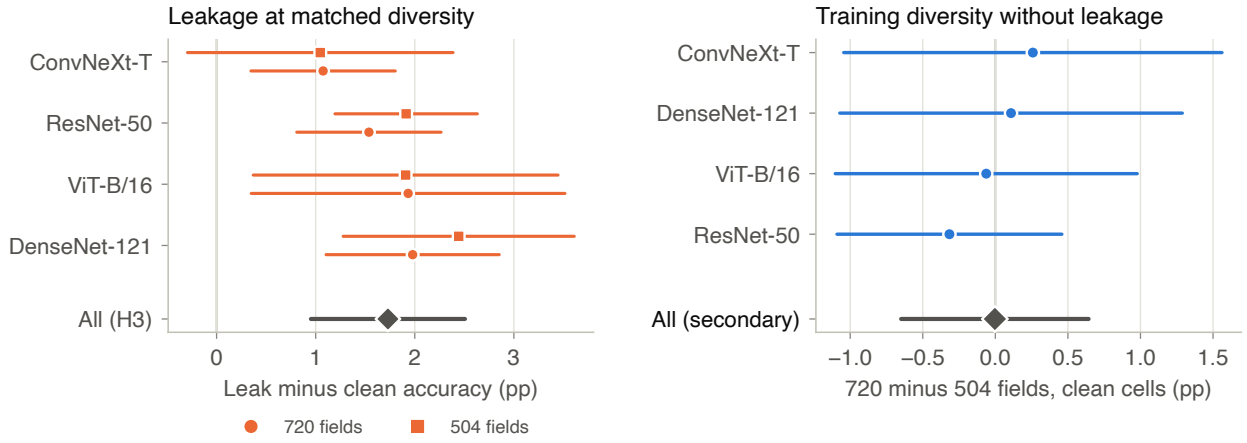


Figure 4.1: Left: leak minus clean accuracy per architecture and diversity level, and the H3 estimate. Right: high minus low diversity within clean cells.

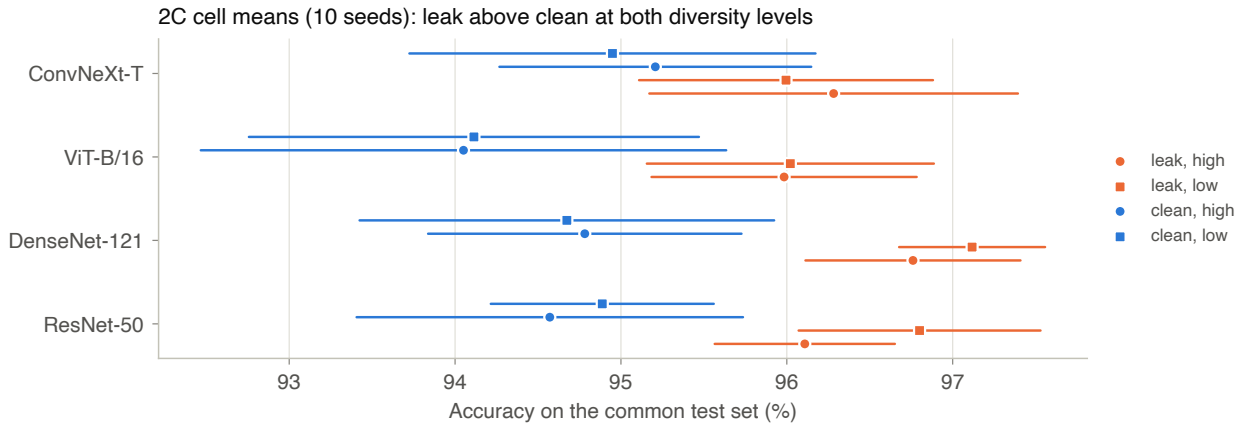


Figure 4.2: All four 2C cell means per architecture (10 seeds each).

The diversity manipulation did change what the training sets contain. In the embedding space of a retrained parent-probe encoder (which identified held-out fields at 24.1% against 0.24% chance), the mean distance between field centroids was 0.499 for 720-field and 0.479 for 504-field clean training sets, and 0.472 (P1, 908 fields) against 0.466 (P2, 773 fields) for the natural splits. More diverse training sets were measurably more diverse; that diversity simply did not improve accuracy.

5

H4: background transplants do not isolate the route

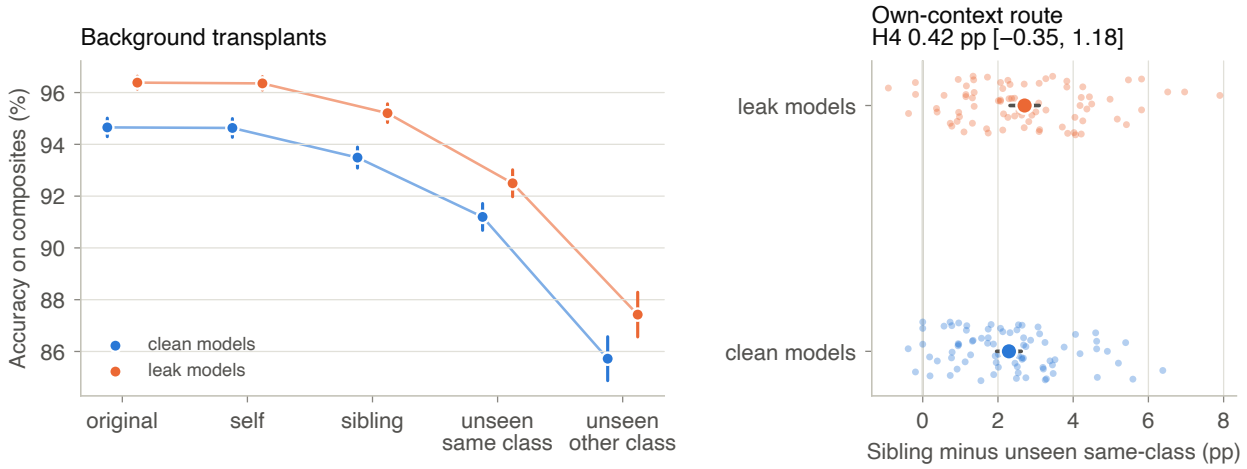


Figure 5.1: Left: accuracy of leak and clean models on composites of the common test set. Right: the per-run own-context route and the registered H4 contrast.

Compositing a cell onto its own inpainted background cost nothing, so seam and inpainting artifacts were negligible. Leak models were 1.73 pp more accurate than clean models on unmodified cells and kept most of that advantage when the background was replaced: 1.72 pp on sibling backgrounds, 1.30 pp on backgrounds from unseen fields of the same class, and 1.70 pp on other-class backgrounds. H4, the leak-minus-clean difference of the own-context route, was 0.42 pp (−0.35 to 1.18; Holm $p = 0.23$): not supported. Its point estimate would assign 24% of the inflation to the background.

Two observations qualify the design. Sibling backgrounds helped even clean models, which had never seen those fields, by 2.29 pp, presumably because a background from the cell's own field matches its stain; that effect is common to both arms but inflates the variance of the contrast. And backgrounds do carry class information: an other-class background moved 3.09 pp of probability toward its class (1.04–5.15, secondary). Restricting H4 to recipients whose background covers at least 30% of the crop gave 0.63 pp (−0.07 to 1.33).

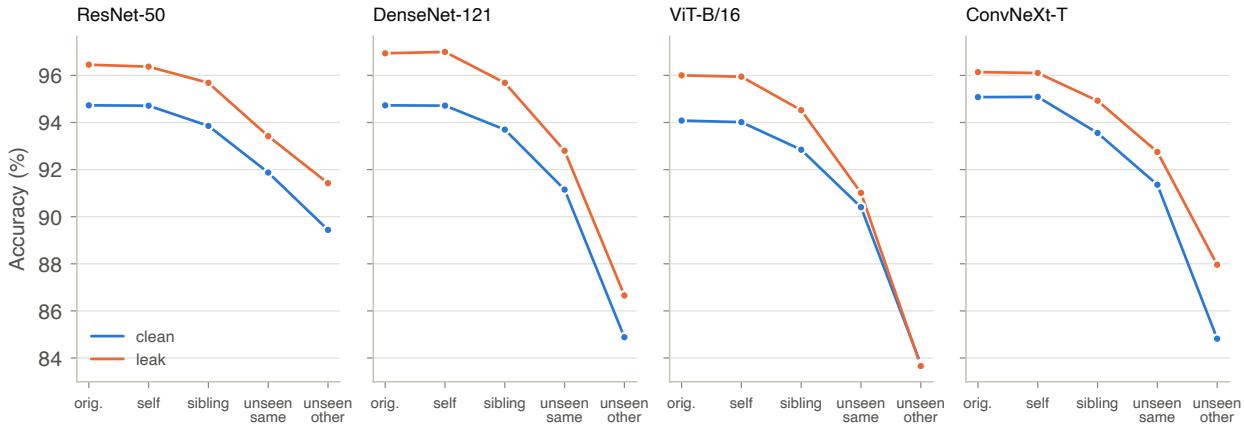


Figure 5.2: Transplant-condition accuracy of leak and clean models per architecture.

Table 5.1: 2C/2B (10 seeds per cell): accuracy on the common test set and on each background-transplant condition (percent), and the donor-class probability shift (pp).

Architecture	Cell	Original	Self	Sibling	Unseen same	Unseen other	Shift
ResNet-50	leak_high	96.11 ± 0.76	95.96 ± 0.77	95.29 ± 0.89	92.61 ± 1.41	90.89 ± 1.24	1.19
	clean_high	94.57 ± 1.63	94.50 ± 1.78	93.43 ± 2.06	91.73 ± 1.94	89.03 ± 2.49	1.35
	leak_low	96.80 ± 1.02	96.78 ± 1.03	96.08 ± 1.03	94.23 ± 1.40	91.95 ± 1.50	1.33
	clean_low	94.89 ± 0.94	94.93 ± 0.83	94.28 ± 0.84	92.02 ± 1.20	89.85 ± 1.16	1.22
DenseNet-121	leak_high	96.76 ± 0.91	96.89 ± 0.77	95.67 ± 0.60	92.80 ± 1.51	86.79 ± 1.96	4.15
	clean_high	94.78 ± 1.32	94.78 ± 1.26	93.72 ± 1.20	90.97 ± 1.99	84.58 ± 2.57	4.50
	leak_low	97.12 ± 0.62	97.10 ± 0.65	95.70 ± 1.28	92.80 ± 2.21	86.52 ± 2.90	4.36
	clean_low	94.67 ± 1.75	94.65 ± 1.83	93.68 ± 1.63	91.33 ± 2.28	85.19 ± 3.46	4.08
ViT-B/16	leak_high	95.98 ± 1.12	95.93 ± 1.22	94.55 ± 2.45	90.44 ± 2.89	84.05 ± 3.05	3.66
	clean_high	94.05 ± 2.21	94.05 ± 2.18	92.82 ± 2.27	90.73 ± 2.43	84.62 ± 4.51	3.36
	leak_low	96.02 ± 1.21	95.96 ± 1.25	94.49 ± 1.36	91.58 ± 2.25	83.26 ± 3.68	4.39
	clean_low	94.11 ± 1.90	93.98 ± 1.97	92.86 ± 2.61	90.07 ± 3.20	82.87 ± 3.39	3.93
ConvNeXt-T	leak_high	96.28 ± 1.55	96.25 ± 1.63	95.01 ± 1.34	92.69 ± 1.81	87.79 ± 3.44	2.71
	clean_high	95.21 ± 1.31	95.19 ± 1.22	93.89 ± 1.32	91.83 ± 2.40	85.63 ± 2.63	3.03
	leak_low	96.00 ± 1.24	95.96 ± 1.17	94.84 ± 2.61	92.81 ± 3.30	88.13 ± 3.33	2.81
	clean_low	94.95 ± 1.71	94.99 ± 1.66	93.22 ± 2.09	90.89 ± 2.82	84.00 ± 4.02	3.40

6

H5: gradient reversal was evaded

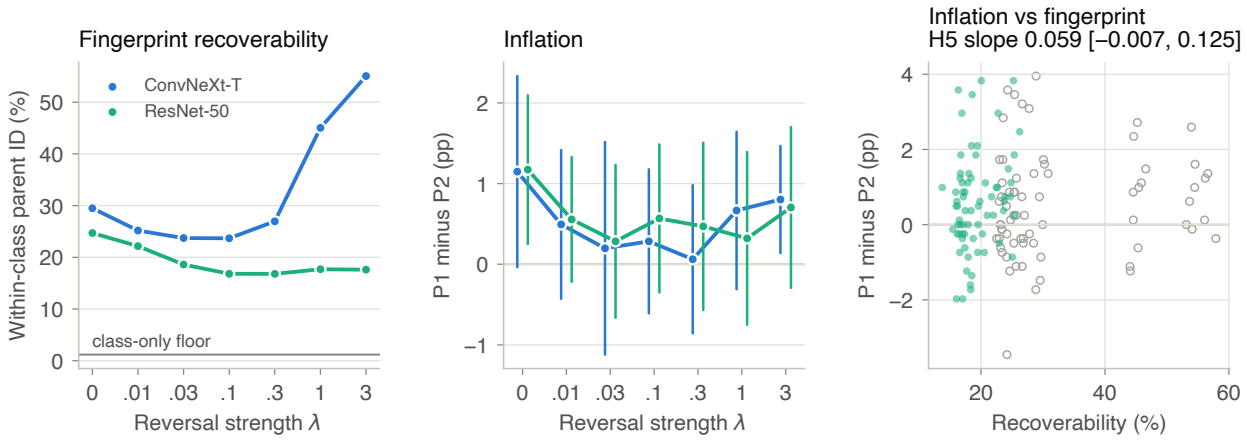


Figure 6.1: The gradient-reversal sweep: probe recoverability, the P1 − P2 gap, and their relation (ConvNeXt-T, which failed its manipulation check, hollow).

For ConvNeXt-T the adversary was evaded: as the reversal strength grew, an independent linear probe identified the field *better* (29.5% at $\lambda = 0$, 55.1% at $\lambda = 3$), while the adversary's own training accuracy fell toward chance (Figure 6.2). The encoder learned features its two-layer adversary could not read but a standardised linear probe could, the failure mode described for adversarial removal in language models [1]. ConvNeXt-T therefore failed the registered manipulation check and was excluded. For ResNet-50, recoverability fell from 24.7% to 17.6% and the mean gap fell from 1.17 pp to between 0.28 and 0.70 pp at every $\lambda > 0$, with P2 accuracy within 1 pp. The registered slope was positive but not significant (0.06 pp per pp, −0.01 to 0.12; Holm $p = 0.16$). Amendment 8's linear adversary later showed the ResNet-50 gap does not fall when recoverability is suppressed on fresh seeds (Part VI), so this drop was most likely noise.

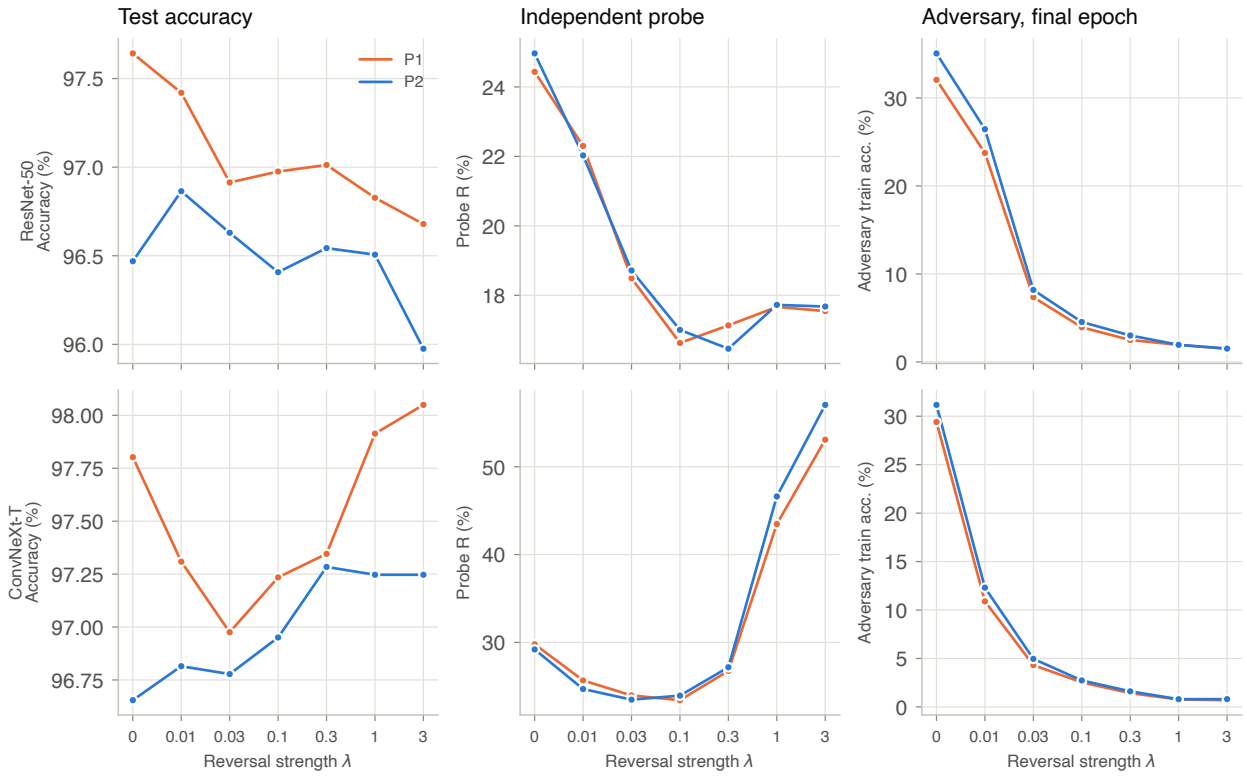


Figure 6.2: The full 2A sweep per architecture: test accuracy, independent-probe recoverability, and the adversary's own within-class training accuracy at the final epoch.

Table 6.1: 2A gradient-reversal sweep (10 seeds): accuracy (percent), mean gap (pp), and within-class parent recoverability R of the frozen features (percent; sklearn probe).

Architecture	λ	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.64 ± 0.46	96.47 ± 1.15	1.17	24.4 ± 1.2	25.0 ± 1.3
	0.01	97.42 ± 0.56	96.86 ± 0.71	0.56	22.3 ± 1.1	22.0 ± 2.0
	0.03	96.91 ± 0.82	96.63 ± 0.84	0.28	18.5 ± 1.7	18.7 ± 1.5
	0.1	96.98 ± 0.38	96.41 ± 1.00	0.57	16.6 ± 1.1	17.0 ± 1.6
	0.3	97.01 ± 0.78	96.54 ± 0.93	0.47	17.1 ± 2.0	16.5 ± 1.0
	1	96.83 ± 0.76	96.51 ± 1.00	0.32	17.7 ± 1.5	17.7 ± 1.7
	3	96.68 ± 0.73	95.98 ± 1.22	0.70	17.5 ± 1.8	17.7 ± 1.5
ConvNeXt-T	0	97.80 ± 0.55	96.65 ± 1.31	1.15	29.8 ± 1.2	29.2 ± 1.5
	0.01	97.31 ± 0.95	96.81 ± 0.73	0.49	25.7 ± 1.3	24.7 ± 1.5
	0.03	96.98 ± 1.00	96.78 ± 1.30	0.20	24.0 ± 1.4	23.5 ± 1.4
	0.1	97.23 ± 0.78	96.95 ± 0.81	0.28	23.4 ± 0.9	23.9 ± 1.1
	0.3	97.35 ± 1.01	97.28 ± 1.05	0.06	26.8 ± 1.5	27.2 ± 1.9
	1	97.91 ± 0.68	97.25 ± 0.82	0.67	43.5 ± 1.0	46.6 ± 1.4
	3	98.05 ± 0.47	97.25 ± 0.65	0.80	53.1 ± 2.0	57.0 ± 2.1

7

What Study 2 established

Study 2 in four sentences

Crop-level splitting inflates accuracy, by 0.97 pp on SIPaKMeD across twelve architectures and by 12.08 pp on CRIC. The inflation is leakage: at matched size and diversity, the test fields' siblings alone add 1.73 pp, and diversity alone adds nothing. The size of the inflation follows the dataset's structure, as the closed forms predict. How models use the leak was not settled: the background route was small and uncertain, and the adversary was evaded.

Bibliography

[1] Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21, 2018.